ASSIGNMENT-1

Foundations of Cryptography

Course: BCA

Subject: Cryptography

Last Submission Date: 5-10-2026

Instructions: Answer all questions. Short-answer questions require 2–3 sentences; long-answer questions require detailed explanations with examples where applicable.

Q1. (Short) Define cryptography in your own words. What are the three main goals it tries to achieve?

Q2. (Short) Differentiate between plaintext and ciphertext with a simple example.

Q3. (Long) Explain the concept of cryptography in detail. Discuss why it is considered essential in the modern digital world, giving at least three real-life applications (e.g., banking, messaging apps, e-commerce).

Q4. (Short) What is the difference between encryption and decryption?

Q5. (Long) Describe the two major types of cryptography — Symmetric Key Cryptography and Asymmetric Key Cryptography. Compare them based on key usage, speed, security, and typical use cases. Support your answer with examples of algorithms under each type (e.g., AES, DES for symmetric; RSA, ECC for asymmetric).

Q6. (Short) What is hashing, and how is it different from encryption?

Q7. (Short) Critically evaluate why achieving confidentiality, integrity and non-repudiation together is challenging in a real-world communication system.

Q8. (Long) Explain the role of cryptography in network security. Discuss how it helps achieve confidentiality, integrity, authentication, and non-repudiation, with one example for each.

Q9. (Short) Name any two network security protocols that use cryptographic techniques and briefly state their purpose (e.g., SSL/TLS, IPSec).

Q10. (Long)Critically discuss the evolution of cryptography from classical ciphers to modern cryptographic systems, evaluating the impact of computational advances on cryptographic security.

Q11. (Long) Trace the historical development of cryptographic techniques, starting from classical methods (like the Caesar Cipher and Vigenère Cipher) through to modern computer-based cryptography (like DES, AES, RSA). Highlight key turning points such as World War II cryptography (Enigma machine) and the invention of public-key cryptography.

Q12. (Short) Who introduced the concept of public-key cryptography, and in which year?